



Cardioversion Procedure

1. Scope

Site	Operational Area	Applicable to
Armada Kalamunda Group	All Clinical Services	Nursing, Midwifery and Medical

2. Purpose

To ensure appropriate procedural care is provided to patients undergoing a cardioversion.

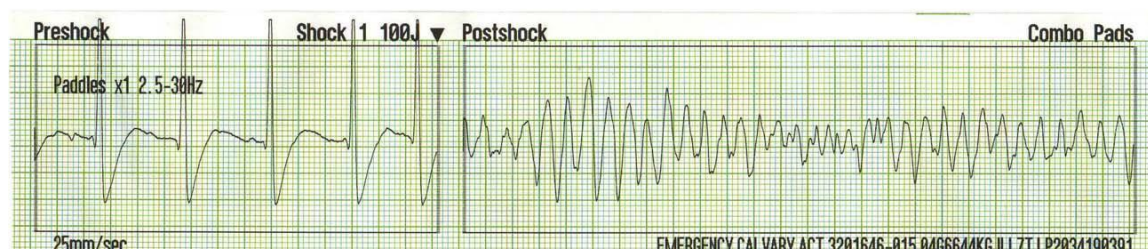
Cardioversion is only to be performed by staff assessed as defibrillator competent.

Nurses who are defibrillator competent may perform cardioversion only under direct supervision of a medical practitioner.

3. General information

Synchronised cardioversion is shock delivery that is timed (synchronised) with the QRS complex. This synchronisation avoids shock delivery during the relative refractory period of the cardiac cycle, when a shock could induce ventricular fibrillation (VF). Known as the R-on-T phenomenon

Figure1: Pre and post shock – discharge not synchronised



4. Background

Transient delivery of electrical current causes a momentary depolarisation of most cardiac cells. This allows the sinus node to resume normal pacemaker activity. Cardioversion is an **elective** or **urgent** procedure used to correct tachyarrhythmias (such as atrial tachycardia, atrial flutter, atrial fibrillation, and symptomatic VT). It is also the treatment of choice for patients with arrhythmia's that don't respond to drug therapy.

5. Procedures

5.1 Patients that may require cardioversion

- Elective outpatient
- Inpatient with new/existing arrhythmia who are haemodynamically stable or unstable.

Indications for elective cardioversion:

- Haemodynamically stable patients to electively restore sinus rhythm (SR) from atrial fibrillation (AF), atrial flutter or supraventricular tachycardia (SVT)
- Stable monomorphic ventricular tachycardia (VT) that does not respond to intravenous anti arrhythmia or electrolyte correction medications.

Indication for Urgent cardioversion Patient with any of the following life threatening features:

1. Shock
2. Syncope
3. Myocardial Ischemia
4. Severe heart failure

5.2 Pre-Procedure Investigations

- The presence of atrial or ventricular thrombus must be excluded if the arrhythmia is suspected of being present for greater than 48 hours and the management is non-urgent. Patients may be anti-coagulated for 4-6 weeks prior to cardioversion and must have a therapeutic international normalisation ratio (INR) prior to cardioversion. Anticoagulation reduces the risk of thromboembolic events during cardioversion.
- Patients with arrhythmia less than 48 hour duration may not require anticoagulation therapy pre-procedure. If arrhythmia more than 48 hour duration, Medical Officer (MO) may elect to perform Transoesophageal Echo (TOE) guided cardioversion without anticoagulant therapy.
- 12 lead ECG.
- Transthoracic Echo.
- Confirm patient pregnancy status. Inform MO if patient is pregnant or pregnancy suspected. Biochemistry e.g. urea, creatinine, U&E, Mg, CK, Troponin, specific drug levels, INR and coagulation. Antiarrhythmic drugs / abnormal electrolyte levels can alter defibrillation threshold.
- In urgent cardioversion, the clinical status of the patient will determine if time permits for the above investigations.
- Consider need for Holter monitor as an outpatient.

5.3 Fasting

- Fast patient as per [AKG Pre-Operative Fasting Guideline](#) or as advised by MO
- For urgent cardioversion, patient may not have been fasted – inform MO and Duty Anaesthetist
- Medications to promote gastric emptying may be prescribed

5.4 Equipment

- Manual defibrillator and resuscitation equipment: checked and functioning
- Oxygen and suction equipment: checked and functioning
- Check airway management equipment is available, including:
 - Simple airway adjuncts (nasal and / or oropharyngeal)
 - Bag-Valve-Mask and tubing is complete and functioning
- Cardiac monitor (for continuous ECG rhythm monitoring)
- Pulse oximetry

- Blood Pressure measuring equipment (Manual or invasive monitoring equipment)
- Intravenous cannulation equipment (including large bore IV cannula)
- Intravenous (IV) sedation and/or analgesia and anaesthetic agents as prescribed by MO on Medication Chart:
 - A short acting IV anaesthetic agent is usually given as DC shocks during cardioversion should never be given to a conscious patient.
 - May have a running IV line of Normal Saline 0.9% via an appropriate giving set, attached to patient.
 - Deliver rate as per the MO order.
- 12 Lead ECG machine available (The 12 lead ECG machine should not be attached to the patient during cardioversion)
- Capnography (EtCO₂), FiO₂ equipment as requested

5.5 Preparation of Patient

The nurse will:

- Ensure consent form is completed.
- Confirm patient has correct identification bands in-situ.
- Place patient in patient gown that allows easy access to chest.
- Clip any excessive hair from chest to improve adherence of defibrillation pads to skin.
- Remove all jewellery as metal jewellery may conduct electrical current resulting in a burn to the patient.
- Encourage patient to void pre procedure so as to reduce the risk of incontinence while under the effect of anaesthesia.
- Dentures are always to remain with the patient.
- Remove if ill-fitting or as per anaesthetic / MO instructions.
- Ensure patient/ carer/significant other (if applicable) understands the procedure by providing verbal information and any written advice that is available.
- Ensure patient's bed/procedure trolley is dry and moisture is dried from chest. (reduces risk of electrical burns and transfer of electrical energy to other personnel)
- Perform and document baseline pre procedure physiological observations on adult observation and response chart. (AKMR 140.3)
- Perform a 12 lead ECG on admission.
- MO to advise regarding withholding of cardio-active drugs e.g. anti- arrhythmic and antihypertensive drugs.
- Elective patients must have fasted as per respective AHS sites policies or as advised by MO.
- Apply 3 or 5 lead cardiac monitoring electrodes to patient's chest, ensuring electrodes are not in contact with defibrillator pads,
- **Note:** If reading waveform via pads, lead placement is not essential.
- Defibrillator pads should not be in contact with medication patches, jewellery, Intra Venous or Central Venous Catheter tubing, dressings and suture lines, and need to be a minimum of 8cm from implantable devices.

- Place defibrillator pads on patient prior to administration of anaesthetic, in anterior-lateral or anterior-posterior position.
 - Refer to Appendix: Defibrillation Pad Placement.

5.6 Cardioversion

- The amount of energy used to cardiovert a patient is not as high as used in emergency situations.
- It is dependent on the arrhythmia being treated or preference of the MO.
- Switch power on.
- Set defibrillator to manual mode.
- Press synchronise (SYNC) button.
- Ensure synchronise function is active by confirming presence of indicator above each QRS complex.
- If defibrillator fails to sense the QRS complex when synchronised cardioversion is performed, the device will not discharge the electrical energy.
- Confirm with MO patient is adequately sedated and prepared for procedure.
- Set and confirm energy level as per MO instructions.
- Press charge and wait for 'ready' display / tone.
- Ensure no personnel have contact with patient or bed and state 'stand clear' prior to discharging shock
- Ensure free flowing oxygen is a minimum of one metre from the defibrillation pads.
- Reconfirm rhythm prior to discharging to deliver electrical current to patient.
- Press and hold 'shock' button until the shock has been delivered.
- Confirm the rhythm after the shock has been delivered.
- Administer further DC shocks as required or prescribed by the MO.
- If cardioversion is unsuccessful, liaise with MO for further management.
- MO may prescribe intravenous anti-arrhythmic medication therapy and may reattempt cardioversion if patient is significantly haemodynamically compromised.
- If sinus rhythm is restored the procedure is concluded.
- If repeat cardioversion is necessary, the procedure can be repeated as determined by the MO.
- Assess patient after each shock is delivered, including airway, breathing and circulation.
- If deterioration of patient condition is noted:

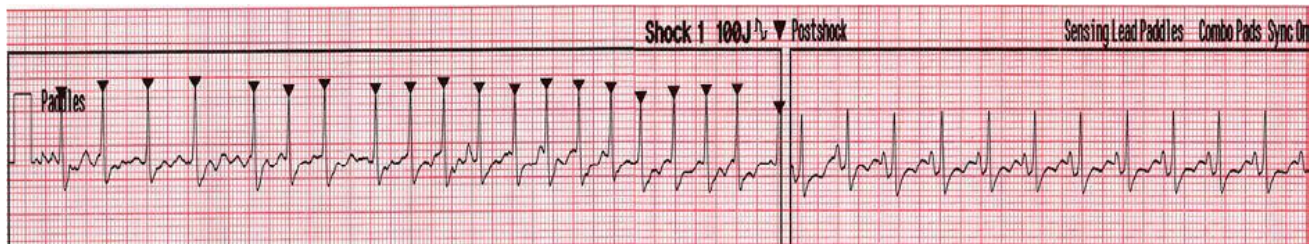
- Follow ALS algorithm.
- If patient requires emergency defibrillation, ensure synchronisation button is switched off.
- Activate MET / resuscitation 'Code Blue' as appropriate
- Provide respiratory support until patient able to maintain own airway and breathing.

5.7 Trouble Shooting

- Increase the QRS amplitude on the monitor, using the SIZE button, (be certain that this does not cause the T-wave to be sensed as the QRS complex)

- Use the LEAD button to select a lead with higher QRS amplitude

Figure: Pre and post shock – arrows indicate synchronisation to R-wave



5.8 Post Procedure

- Assess the patient for complications post procedure.
- Hypoxia / hypoventilation from sedation
- Partial thickness burns.
 - Avoid the use of lignocaine gel or creams on burns as lignocaine gel has the potential to induce serious toxicity (anti arrhythmic) with systemic absorption.
 - Creams/gels can leave a thin film on skin surface, making it more difficult for application of self - adhesive electrodes and increased thoracic impedance if further cardioversion is required.
- Cardiac arrhythmias.
- Nursing staff to liaise with the MO if complications present and document on appropriate chart / notes.

5.9 Observations

- Conscious state.
- Respiration rate and SpO2.
- Cardiac rhythm and pulse.
- Blood pressure.
- Evidence of pain and / or skin burns.
- Obtain post procedure ECG and compare with pre procedure ECG.
- In Emergency Department (ED), record full neurological observations.
- Assess and report pain and nausea.
- Document on Adult observation and response chart (AKMR140.3).
- Offer the patient diet and fluids post procedure if asymptomatic, patient awake and gag reflex present.
- Provide the patient with routine discharge advice following an anaesthetic agent
- Report directly to cardiologist and anaesthetist or Senior treating MO in ED if any problems are noted or initiate code blue if appropriate.

5.10 Frequency of Observations

- 5 / 60 until patient regains consciousness.
- 15 / 60 and prn for 30 minutes.

- 30 / 60 and prn for 1-2 hours.
- 1 / 24 or as clinically indicated.

6. Legislative context

The following documents are required to give effect to this procedure:

- Health Department of Western Australia (HDWA) [Clinical Handover Policy MP 0095](#)
- HDWA [Recognising and Responding to Acute Deterioration Policy MP 0171/22](#)

7. Supporting information

The following documents support the implementation of this procedure:

- [Electrocardiogram \(ECG\) – 12 Lead ECG Recording and Review Procedure AKG-PRO-0145-11](#)
- [Pre-operative Fasting Guideline THE-GUI-0369-16](#)
- East Metropolitan Health Service (EMHS) [Inter-Hospital Patient Transfer EMHS: 38](#)
- EMHS [Appendix A IHPT Roles, Responsibilities and Escalation](#)

8. Relevant standards

National Safety and Quality Health Service (NSQHS) Standards (second edition):

- Standard 5 – Action 5.28 Nutrition and hydration
- Standard 6 – Action 6.11 Documentation of information
- Standard 8 – Action 8.4 Recognising acute deterioration
- Standard 8 – Action 8.6 Escalating care

9. References

1. Cardioversion Clinical Practice Standard SMSH Code: CPSC001
2. Cardioversion Management Elective and Urgent Nursing Practice Standard RPH
3. Cardioversion Management of Elective and Urgent Nursing Practice Standard FSH

10. Glossary

The following definitions are relevant to this policy.

Term	Definition
Cardioversion	Cardioversion is an elective or urgent procedure used to correct tachyarrhythmia (such as atrial tachycardia, atrial flutter, atrial fibrillation and symptomatic VT). It is also the treatment of choice for patients with arrhythmias that don't respond to drug therapy.
VT	Ventricular tachycardia
AF	Atrial fibrillation
SVT	Supraventricular tachycardia
SR	Sinus rhythm

Term	Definition
VF	Ventricular fibrillation

11. Policy owner (relating to these procedures)

Director of Clinical Services

Enquiries relating to this procedure may be directed to:

Title: Director of Clinical Services
 Department: Clinical Services
 Email: tania.strickland@health.wa.gov.au

12. Review

This procedure will be reviewed and evaluated as required to ensure relevance and currency. At a minimum it will be reviewed within one (1) year after first issue and at least every three (3) years thereafter.

Version	Effective from	Effective to	Amendment(s)
3	18/06/2024	18/06/2027	

13. Authorisation

This procedure has been authorised and issued by the [enter position title of the authoriser].

Authorised by	Neil Cowan – Executive Director
Authorisation date	18/06/2024
Published date	18/06/2024

Appendix 1: Defibrillation pad placement

Defibrillation pads are applied with a rolling motion to prevent air being trapped beneath the electrodes. If the pads touch, electrical arcing may occur resulting in burns to the patient and unsuccessful cardioversion.

Anterior-Lateral Defibrillation Pad Placement

Place pads on the exposed chest in an anterior-lateral position, unless advised otherwise by MO or clinically appropriate to do so. In large breasted individuals it is reasonable to place the left electrode pad lateral to or underneath the left breast avoiding breast tissue

- Left 6th intercostal space mid axilla line
- Right 2nd intercostal space right sternal border
- Defibrillator pads must not be touching each other.

Figure : Cardioversion Anterior-Lateral Pad Placement



Anterior-Posterior Placement

The anterior-posterior placement is preferred in patients with implantable devices at least 8cm away, in order to avoid shunting current to the implantable device and damaging the system. The MO may prescribe anterior-posterior pad placement if patient has a history of difficult previous cardioversion, previous requirement for high-energy joules for cardioversion or obesity/ large body size.

Low energy joules are used where possible, to avoid damage to the implantable device or PPM. The implantable device or PPM must be checked post cardioversion by a cardiac technician.

- Anterior pad – 4th-5th intercostal space, mid-clavicular line
- MO may request variation of anterior pad to higher placement
- Posterior pad – Place under left scapula at the level of the heart

Figure: Cardioversion Anterior-Posterior Pad Placement

